

System Design Interview Question Bank

Difficulty split: Easy / Medium / Hard

Easy

1. Design a URL shortener for internal company use.
2. Design a simple pastebin-like text sharing service.
3. Design a rate limiter for a public API.
4. Design file upload + download service for profile images.
5. Design a basic notification service (email + push).
6. Design a recently viewed items feature for an e-commerce app.
7. Design a search autocomplete service for product names.
8. Design a URL redirection tracking service (count clicks, top links).

Medium

1. Design a URL shortener at internet scale (high read traffic, analytics).
2. Design a chat service (1:1 + group, online/offline delivery).
3. Design a distributed job queue system (retries, dead-letter queue).
4. Design a real-time collaborative document commenting system.
5. Design an event ingestion pipeline for app logs and metrics.
6. Design a recommendation feed service for a social app.
7. Design an API gateway for multiple microservices (auth, throttling, routing).
8. Design a booking system (seat/slot locking, concurrency, cancellation).

Hard

1. Design YouTube-like video upload, processing, and streaming platform.
2. Design Uber-like ride matching system with real-time location updates.
3. Design a global news feed system (fan-out tradeoffs, ranking, freshness).
4. Design a multi-region payment processing system (idempotency, consistency, fraud checks).
5. Design a globally distributed key-value store (consistency vs availability).
6. Design a real-time bidding ad platform (low latency + high throughput).
7. Design a cloud object storage service (S3-like) with durability guarantees.
8. Design a large-scale observability platform (logs, metrics, traces).

Expected Signals (Evaluation Rubric)

1. Requirement clarification quality
2. Data model choices
3. API contract clarity
4. Scaling strategy (partitioning/sharding/caching)
5. Reliability (retry, failover, backpressure)
6. Consistency tradeoff reasoning
7. Security and abuse prevention
8. Monitoring and operational readiness